Donghyun Kim

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Research Interests

3D Vision: Point Cloud Processing, 3D Reconstruction, Point Cloud Restoration.

Human-Centric Vision: Human Performance Capture, Novel-View Human Animation, 3D Human Pose Analysis, Video-Based Inverse Dynamics.

Education

Yonsei University — **Seoul, South Korea**
Ph.D. in Computer Science (Advisor: Prof. Seong Jae Hwang) — Sep 2023 – Present

Yonsei University — **Seoul, South Korea**
B.S. in Computer Science — Mar 2019 – Aug 2023

Publications

Point Clouds in the Frequency Domain: Color in Amplitude, Geometry in Phase
Donghyun Kim, Chanyoung Kim, Junhyeok Kim, Eunseo Jeong, Hyunah Ko, Youngjoong Kwon, Seong Jae Hwang
Under Review, 2026.

Delaunay Canopy: Building Wireframe Reconstruction from Airborne LiDAR Point Clouds via Delaunay Graph
Donghyun Kim, Chanyoung Kim, Youngjoong Kwon, Seong Jae Hwang
ECCV 2026 [Paper] | [Project]

How Noisy Poses Break Inverse Dynamics: Analysis and Mitigation for Video-Based Joint Torque Estimation
Donghyun Kim, Chanyoung Kim, Eunseo Jeong, Youngjoong Kwon, Seong Jae Hwang
CVPRW 2026 (PhysHuman) [Paper]

Rethinking Graph Convolution for 2D-to-3D Hand Pose Lifting
Chanyoung Kim, Donghyun Kim, Dong-Hyun Sim, Seong Jae Hwang, Youngjoong Kwon
CVPRW 2026 (PhysHuman) **Oral Presentation** [Paper]

FALCON: Fast Image Haze Removal Leveraging Continuous Density Mask
Donghyun Kim, Seil Kang, Seong Jae Hwang
CVPRW 2025 (NTIRE) [Paper]

PLATYPUS: Progressive Local Surface Estimator for Arbitrary-Scale Point Cloud Upsampling
Donghyun Kim, Hyeonkyeong Kwon, Yumin Kim, Seong Jae Hwang
AAAI 2025 **Oral Presentation (Top 4.6%)** [Paper] | [Project]

WoLF: Wide-scope Large Language Model Framework for CXR Understanding
Seil Kang, Donghyun Kim, Junhyeok Kim, Seong Jae Hwang
arXiv preprint, 2024. [Paper]

Research & Industry Experience

Emory University (Emory Vision and Graphics Lab) — **Atlanta, GA, United States**
External Collaborator — Jul 2026 – Present
Visiting Researcher (Advisor: Prof. Youngjoong Kwon) — Feb 2026 – Jun 2026
– Ongoing projects on Human Performance Capture and Novel-View Human Animation

Arizona State University (ASU) **Tempe, AZ, United States**
Short-Term Scholar Feb 2024

Yonsei University (Medical Imaging and Computer Vision Lab) **Seoul, South Korea**
Ph.D. Student (Advisor: Prof. Seong Jae Hwang) Sep 2023 – Present
Undergraduate Student Researcher Jul 2022 – Aug 2023
– Point Cloud Processing, 3D Reconstruction, Point Cloud Restoration, Image Dehazing, CXR Understanding

Innoyard **Seoul, South Korea**
Researcher & Software Engineer Jun 2021 – Aug 2022
– Project "InnoScan": Mobile 3D human body data capturing service using 3D point cloud registration and surface reconstruction

Yonsei University (Dependable Computing Lab) **Seoul, South Korea**
Undergraduate Student Researcher (Advisor: Prof. Kyoungwoo Lee) Jan 2021 – Jun 2021
– Project "FluidTrack" (with Severance Children's Hospital): Child-parent collaborative tracking system for pediatric voiding dysfunction

🏛 Scholarly Activities & Services

Program Committee
ECCV 2026 MoCha Workshop

Conference Reviewer
CVPR (2026 **Outstanding Reviewer**), NeurIPS (2026 **Top Reviewer**), ECCV (2026), ICCV (2025), WACV (2024, 2025), 3DV (2026)

Journal Reviewer
IEEE Transactions on Image Processing (TIP), Pattern Recognition (PR)

Teaching Assistant
Yonsei University (Introduction to Deep Learning and Application) Spring 2024, Spring 2025, Spring 2026

Invited Lecturer
Samsung Electronics (Introduction to Data Science) Sep 2023 – Oct 2023

🏆 Honors & Awards

Graduation Project **Grand Prize** Jun 2023
Yonsei University

Graduation Project **Best Demo Award** Dec 2022
Yonsei University

Coding Challenge **2nd Prize** Dec 2021
Korea Institute of Startup & Entrepreneurship Development (KISED)